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Chapter 5 — Data Visibility: PoC, DMA Interactions, DC CVAC/CIVAC Semantics, and Barrier Requirements (Based on ARM64 Cortex-A72)

5.1 Scope

This chapter defines the architectural and microarchitectural mechanisms governing **data visibility** on ARM64 Cortex-A72. It focuses on:

- The Point of Coherence (PoC)
- DMA interactions
- DC CVAC and DC CIVAC semantics
- Dirty-line propagation
- Snoop behavior
- Required barrier usage
- Correct sequences for CPU→DMA and DMA→CPU communication

This chapter complements Chapter 4 (instruction visibility) by addressing the data path.

5.2 Data Visibility Domains

Data visibility is governed by three architectural domains:

5.2.1 L1D — Core-Local Data Cache

- Private to each core
- Write-back
- Contains dirty lines
- Not visible to DMA
- Not visible to other cores

5.2.2 L2 — Cluster-Shared Unified Cache (PoU)

- Shared by all cores in the cluster
- Coherent within the cluster
- May contain dirty lines
- Not visible to DMA
- Serves as PoU for instruction visibility

5.2.3 PoC — Point of Coherence

- System-wide visibility domain
- Observed by DMA engines, other clusters, and DRAM controller
- Architecturally defined as the domain where all agents observe consistent memory

PoC is the **only domain visible to DMA**.

5.3 Dirty Line Semantics (Cortex-A72)

Dirty lines may exist in:

- L1D
- L2

Dirty lines represent memory that:

- Has been modified by the CPU
- Has not yet been written back to PoC
- Is invisible to DMA and other cores outside the cluster

Dirty lines must be explicitly propagated using DC CVAC or DC CIVAC.

5.4 Snoop Behavior (Cortex-A72)

Cortex-A72 implements:

- **Duplicate directory snoops**
- L2 snoops L1D for dirty lines
- Snoop responses propagate through FEQ (Fill/Evict Queue)
- Snoop completion is guaranteed only after **DSB**

L1I is **not** snooped.

5.5 DC CVAC — Clean to Point of Coherence (PoC)

5.5.1 Purpose

Push dirty data from L1D/L2 to PoC.

5.5.2 Architectural Effect

- Ensures DMA reads updated data
- Ensures external agents observe correct memory
- Required before CPU→DMA communication

5.5.3 Cortex-A72 Behavior

- L2 snoops L1D for dirty lines
- Dirty lines are written back to PoC
- FEQ pipelines the writeback
- Completion requires **DSB ISH**

5.6 DC CIVAC — Clean + Invalidate to PoC

5.6.1 Purpose

Write back dirty data to PoC and invalidate local copies.

5.6.2 Architectural Effect

- Required after DMA writes to memory
- Ensures CPU refetches fresh data from PoC
- Prevents stale L1D/L2 lines from being used

5.6.3 Cortex-A72 Behavior

- Dirty lines written back
- Clean lines invalidated
- L1D and L2 entries removed
- Completion requires **DSB ISH**

5.7 DC IVAC — Invalidate to PoC

5.7.1 Purpose

Invalidate local cache line without writeback.

5.7.2 Architectural Effect

Used when:

- DMA writes to memory
- The CPU knows the line is clean
- No writeback is required

5.8 Required Barrier Semantics

5.8.1 DSB ISH — Data Synchronization Barrier

Required after:

- DC CVAC
- DC CIVAC
- DC CVAU (when used for data visibility)

DSB ensures:

- All writebacks complete
- All snoops complete
- All FEQ entries drain
- All visibility transitions are architecturally complete

5.8.2 ISB

Not required for data visibility. ISB is only required for instruction visibility.

5.9 Correct Usage Patterns

5.9.1 CPU → DMA (Device Reads CPU Data)

Required sequence:

```
DC   CVAC, xN
DSB  ISH
```

Ensures:

- Dirty lines propagate to PoC
- DMA reads updated memory

5.9.2 DMA → CPU (Device Writes Data)

Required sequence:

```
DC   CIVAC, xN
DSB  ISH
```

Ensures:

- CPU invalidates stale lines
- CPU refetches fresh data from PoC

5.9.3 CPU → External Agent (Other Cluster)

Required sequence:

```
DC   CVAC, xN
DSB  ISH
```

Other clusters observe memory only at PoC.

5.9.4 External Agent → CPU (Other Cluster Writes)

Required sequence:

```
DC   CIVAC, xN
DSB  ISH
```

Ensures CPU refetches updated data.

5.10 MMIO and Device Memory

Device-nGnRnE regions:

- Must not be cached
- Must not use DC/IC operations
- Require DSB/ISB for ordering only
- Are always visible at PoC

Incorrectly caching MMIO results in undefined behavior.

5.11 Microarchitectural Notes (Cortex-A72)

Although hidden behind architectural barriers, Cortex-A72 implements:

- Write-back L1D and L2
- Non-inclusive L2
- FEQ pipelining for writebacks
- Duplicate directory snoops
- Register slices for large L2 configurations

These do not alter architectural semantics.

5.12 Exercises (Technical Reference)

Exercise 5.1 — Dirty Line Propagation

Explain why DC CVAC is required before DMA reads CPU-generated data.

Exercise 5.2 — DMA Write Visibility

Describe why DC CIVAC must be executed after DMA writes to memory.

Exercise 5.3 — PoC Semantics

Explain why PoC is the only domain visible to DMA.

Exercise 5.4 — Barrier Requirements

Describe the architectural role of DSB ISH in completing DC CVAC/CIVAC operations.

Exercise 5.5 — MMIO

Explain why DC/IC operations must not be used on Device-nGnRnE regions.